2023 3rd International Conference on Smart Generation Computing, Communication and Networking (SMART GENCON)
Karnataka, India. Dec 29-31, 2023

2023 3rd International Conference on Smart Generation Computing, Communication and Networking (SMART GENCON) | 979-8-3503-1912-5/23/$31.00 ©2023 IEEE | DOI: 10.1109/SMARTGENCON60755.2023.10441927

Designing and Implementation of 4-bit Carry Look Ahead Adder using CMOS Logic

1st Divya.B
Department of Electronics and Communications
Vignana Bharathi Institute of Technology
Hyderabad, India
beebireddy.divya@gmail.com

2nd Ranisha Koripelly
Department of Electronics and Communications
Vignana Bharathi Institute of Technology
Hyderabad, India
koripellyranisha8688@gmail.com

3rd Harideep.B
Department of Electronics and Communications
Vignana Bharathi Institute of Technology
Hyderabad, India
harideepreddy207@gmail.com

4th Mahesh Killalola
Department of Electronics and Communications
Vignana Bharathi Institute of Technology
Hyderabad, India
killalolamahesh@gmail.com

5th Jyothsna.T
Department of Electronics and Communications
Vignana Bharathi Institute of Technology
Hyderabad, India
joshsujjireddy@gmail.com

***Abstract*—In digital electronics for addition we use full adders but it has high carry propagation delay,while using full adders in the operations like multiplications,divisions and other calculations as the bit size increases carry propagation increases and we have to use complex structures so we will go with carry look ahead adder which will not only produces the successive carries simultaneously but as number of bits increases its propagation delay will be less compared to many adders,here we will design 4-bit carry look ahead adder in cadence tool in gpdk45 technology by cmos technology using basic logic gates like transmission gate, or gate,and gate,ex-or gate [1].**

***Index Terms*—Carry look ahead adder, carry propagation, logic gates, Cmos technology**

I. Introduction

In the realm of digital circuit design, the creation of fundamental logic gates like AND, OR, and XOR forms the bedrock upon which complex com- putational architectures are built. When we combine these elemental logic gates with the versatile transmission gate technology, we embark on a journey to craft a 4-bit Carry Look-Ahead Adder (CLA) within the powerful con- fines of Cadence's design tools.The world of Complementary Metal-Oxide- Semiconductor (CMOS) technology and Cadence presents us with a canvas where we can artistically mold these basic building blocks into an intricate digital masterpiece. CMOS, with its attributes of low power consumption and high performance, is a natural fit for crafting advanced digital circuits [2].

A. Carry propagation reduction in carry look ahead adder

Carry propagation in a Carry Look-Ahead Adder (CLA) is significantly faster than in a Ripple Carry Adder (RCA) due to its parallel processing approach. In a Ripple Carry Adder, the carry bit generated from one stage is dependent on the carry-out from the previous stage, creating a sequential carry propagation. This sequential nature leads to longer critical paths and slower overall performance. In contrast, a CLA employs a more sophisticated technique to compute carry bits in parallel, which reduces the carry propagation delay.

Here's how a Carry Look-Ahead Adder achieves faster carry propagation:

1. Generate (G) and Propagate (P) Signals: In a CLA, each bit pair to be added generates two signals, G (Generate) and P (Propagate). These signals are pre computed based on the input bits and represent the local conditions that determine if a carry is necessary or if it can propagate without interruption.

2. Generate Carry Terms: The G and P signals are used to generate the carry-out for each bit stage using logic gates. The carry-out for each bit is computed independently of the inputs and outputs of other bits, making it a parallel process.

3. Generate Sum Bits: Simultaneously, the sum bit for each stage is computed using XOR gates based on the input bits and the carry-in signal.

4. Parallel Carry Computation: Since the carry-out for each bit is computed in parallel using the G and P signals, there is no sequential dependency on the carry-out of the previous stage. As a result, all carry bits can be determined at the same

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time.

5. Reduced Critical Path: The CLA reduces the critical path of the adder because the carry propagation delay is primarily determined by the logic gates used to calculate the carry-out terms. These logic gates have a fixed delay and are independent of the number of bits in the adder, resulting in a constant carry propagation delay for any input size.

In summary, a Carry Look-Ahead Adder minimizes carry propagation delay by pre computing carry generation conditions (G and P signals) and calculating carry bits for each stage in parallel. This parallelism allows for much faster addition of multi-bit numbers compared to the sequential carry propagation seen in Ripple Carry Adders [3].

B. Equations to build 4-bit carry look ahead adder

A 4-bit Carry Look-Ahead Adder (CLA) combines four 1-bit full adders with additional logic to generate carry signals efficiently. The carry-out from each full adder depends on the inputs and the carry-in signal. The equations for a 4-bit Carry Look-Ahead Adder are as follows: Let A3, A2, A1, A0 be the 4-bit binary inputs, B3, B2, B1, B0 be the 4-bit binary inputs to be added, and C0 in be the carry-in signal. The carry-out C3 (Cout) and sum (S) for each bit are calculated as follows [4].

$$S0 = A0 \oplus B0 \oplus Cin \tag{1}$$

$$C0 = (A0 \bullet B0) + (A0 \bullet Cin) + (B0 \bullet Cin) \tag{2}$$

For the second bit (Bit 1):

$$S1 = A1 \oplus B1 \oplus G1 \tag{3}$$

$$C1 = (A1 \bullet B1) + (A1 \bullet G1) + (B1 \bullet G1) \tag{4}$$

The Generate (G) and Propagate (P) signals for Bit 1:

$$G1 = A1 \bullet B1 \tag{5}$$

$$P1 = A1 \oplus B1 \tag{6}$$

For the third bit (Bit 2):

$$S2 = A2 \oplus B2 \oplus G2 \tag{7}$$

$$C2 = (A2 \bullet B2) + (A2 \bullet G2) + (B2 \bullet G2) \tag{8}$$

The Generate (G) and Propagate (P) signals for Bit 2:

$$G2 = A2 \bullet B2 \tag{9}$$

$$P2 = A2 \oplus B2 \tag{10}$$

For the most significant bit (MSB):

$$S3 = A3 \oplus B3 \oplus G3 \tag{11}$$

$$C3 = (A3 \bullet B3) + (A3 \bullet G3) + (B3 \bullet G3) \tag{12}$$

The Generate (G) and Propagate (P) signals for Bit 3:

$$G3 = A3 \bullet B3 \tag{13}$$

$$P3 = A3 \oplus B3 \tag{14}$$

These equations describe the operation of a 4-bit Carry Look-Ahead Adder, where the carry-out (Cout) of each bit is generated independently based on the Generate (G) and Propagate (P) signals, allowing for faster carry propagation compared to a ripple carry adder [5].

C. Transmission gate

A transmission gate, often referred to as a ”transmission switch” or ”analog switch,” is an electronic component used in digital and analog circuits to control the flow of signals. It is a type of semiconductor switch that allows signals to pass through or be blocked, similar to a mechanical gate controlling the flow of traffic. Transmission gates are widely used in various applications within the field of electronics due to their versatility and efficiency. In this experiment we will use transmission gate to construct the gates like logic and gate,logic or gate,logic not gate and logic ex-or gate.

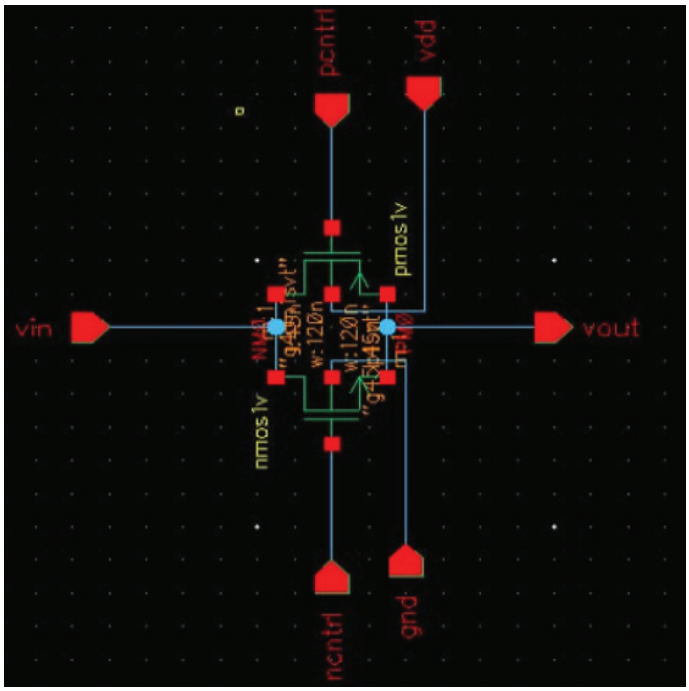


Fig. 1. Transmission gate schematic

II. Experimental setup

In the following experiment we will construct the 4-bit cla using transmission gate logics of or,and,not and ex-or gates to generate sum and carry logics. Here to construct the CLA we will use logic and ,logic or and logic ex-or as sub modules and make wire connectinons as required or given in cla equations pdf,the above paper is different because here we will use transmission gate logic instead of general gates. Fig. 2 shows the schematic diagram of 4-bit carry look ahead adder using basic cmos logic gates like transmission gate,or gate,ex-or gate and logic and gate.

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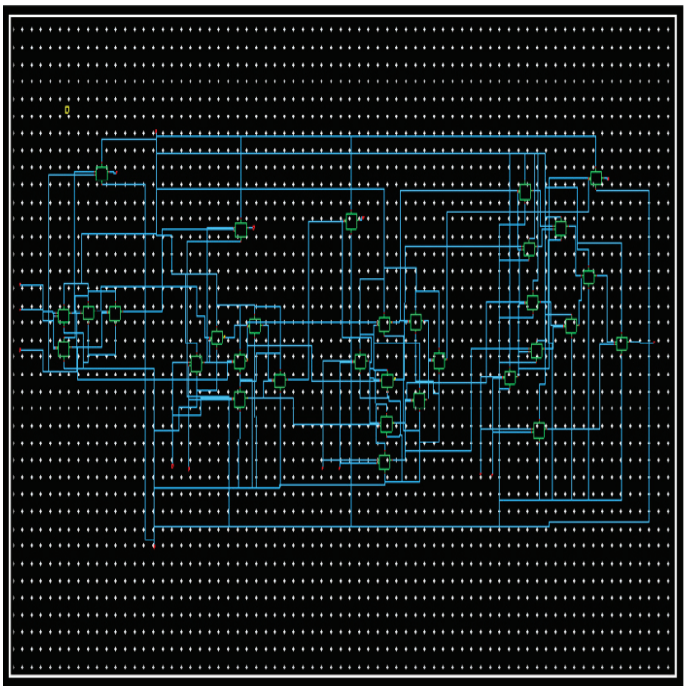


Fig. 2. Schematic of 4-bit carry look ahead adder

III. Results and Analysis

The simulation for the above schematic is done by creating symbol and test schematic and launching the test schematic of 4-bit carry look ahead adder.

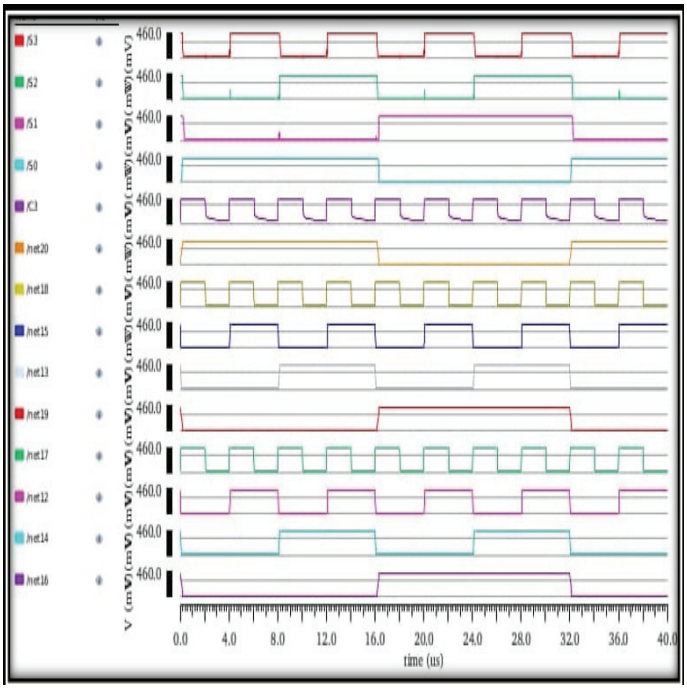


Fig. 3. Simulation of 4-bit carry look ahead adder.

In summary, a CLA offers reduced delay compared to RCAs, especially for larger bit widths, making it a preferred choice for high-speed applications. However, its power consumption is often higher, primarily due to static power, which may influence its suitability for low-power designs [6]. Designers must carefully balance speed and power requirements when choosing a CLA or considering other adder architectures [7].

ADDER	4-BIT CARRY LOOK AHEAD ADDER	4-BIT RIPPLE CARY ADDER
POWER CONSUMPTION	$16.487*10^{-9}$ watts	$765.7*10^{-12}$ watts
DELAY	3.626 nS	21.9 nS

Fig. 4. Comparision between power and carry propagation delay of 4-bit carry look ahead adder and ripple carry adder.

We observe that delay is approximatly 5 times less than the ripple carry adder but power comsuption is very high as we use more number of gates compared to ripple carry adder [8].

IV. Conclusion

In conclusion, designing and implementing a 4-bit Carry Lookahead Adder in Cadence involves a systematic approach that leverages the capabilities of Cadence's design and simulation tools. This process ensures the adder's correctness, performance, and adherence to design constraints, making it suitable for a wide range of digital circuit applications. Cadence's comprehensive tool set empowers designers to create efficient and high-performance digital circuits like the 4-bit CLA.

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